

Smart Face Attendance System Using FaceNet

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Abstract— As organizations transition to digital and contactless systems, automated attendance tracking has grown in significance. Conventional methods, such as card-based identification or manual entry, frequently suffer from mistakes, abuse, or deliberate proxy attendance. To address these limitations, this work introduces a Smart Face Attendance System built on FaceNet, a deep-learning model that represents facial features as compact 128-dimensional embeddings. The system combines MTCNN for reliable face detection and alignment with a lightweight refinement process designed to stabilize embeddings when faces are captured under poor lighting, partial occlusion, or other real-world disruptions. An adaptive thresholding technique modifies the decision boundary based on the observed embedding patterns during operation to further improve verification accuracy. A custom dataset gathered in office and classroom settings was used to test the system, and the results show appreciable gains in recognition consistency, particularly under difficult circumstances. Crucially, only numerical embeddings—rather than unprocessed face images—are saved, enabling deployment that respects privacy. All things considered, the suggested framework provides a workable and expandable solution for contemporary automated attendance systems.

Keywords—FaceNet, MTCNN, automated attendance system, biometric authentication, refined embedding, adaptive thresholding, illumination variation, occlusion robustness, and biometrics that protect privacy.

I. INTRODUCTION

The growing digitalization of educational institutions, workplaces, and public systems has heightened the demand for reliable and secure identity verification solutions. Conventional forms of attendance—such as manual signing, RFID cards, barcodes, and password-based systems—can be subject to proxy attendance, lost or duplicated tokens, and human error. Biometric-based options have gained traction in utilization due to their non-transferability, ease of

implementation, and low user engagement. Among the different modalities available for biometric systems, face recognition is particularly preferable because it is contactless and utilizes existing camera infrastructure.

Earlier face recognition methods that utilized handcrafted features like Eigenfaces, Fisherfaces, and Local Binary Patterns (LBP) were severely impaired in conditions that could not be controlled such as illumination imbalance, pose variation, facial occlusions and expression changes. The advent of deep learning has significantly reduced these barriers by allowing models to learn to directly represent robust discriminative features from image data. A prominent advancement was Google's FaceNet which framed face recognition as a metric learning problem. Rather than classifying faces by identity, FaceNet embed face images into a compact 128-dimensional embedding space that was optimized through triplet loss for intra-class compactness and inter-class separability. This embedding based approach is amenable for scalable verification and is well suited for attendance problems when new users are frequently added during which a model may not always be retrainable.

This work presents a Smart Face Attendance System using FaceNet, designed to provide accurate, real-time, and privacy-preserving attendance automation. Unlike generic face recognition frameworks, the proposed system incorporates a complete operational pipeline comprising camera-based image acquisition, MTCNN-based face detection and alignment, FaceNet embedding generation, a novel embedding refinement strategy for illumination and occlusion robustness, adaptive Euclidean distance-based similarity evaluation, and secure attendance logging. The introduced refinement mechanism stabilizes embedding vectors under inconsistent environmental conditions, thereby enhancing practical deployment reliability in classrooms and workplace settings.

II. RESEARCH SYNTHESIS

The domain of face recognition has evolved from simple geometric feature extraction to highly efficient deep learning-based embedding systems. More classical methods like Eigenfaces, Fisherfaces, and LBP mostly consider linear feature projections or texture-based descriptors. These methods, however, were lightweight in computation but

couldn't generalize well beyond constrained datasets due to real-world pose, illumination, and facial expression changes..

A new era in automatic facial representation learning emerged with the advent of Convolutional Neural Networks. CNN-based systems can learn hierarchical and abstract features directly from image pixels, effectively outperforming the limitation of handcrafted features. Pioneering works regarding modern face recognition frameworks were reported by DeepFace from Facebook (Taigman et al., 2014) and VGGFace from Oxford's Visual Geometry Group (Parkhi et al., 2015). DeepFace leveraged a nine-layer neural network trained on four million facial images to achieve human-like performance on the Labeled Faces in the Wild dataset. Likewise, focusing on large-scale training and deep architecture, the VGGFace data achieved impressive accuracy through the learning of supervised classifications.

While these approaches achieved good performance, early CNN-based methods were still identity-classification-centric, predicting discrete identity labels in the final layer. Most of those methods required retraining once new individuals had been added to the database, which, for scalable deployment, was computationally inefficient. Besides, their learned features lacked a direct measure of facial similarity, restricting adaptability in verification and clustering applications.

Following FaceNet, several works proposed various improvements concerning computational efficiency, domain adaptation, and fairness. Representative works include SphereFace, ArcFace, and CosFace, which introduced angular margin-based losses to enhance the inter-class separability in the embedding space. Others explored lightweight architecture designs, such as MobileFaceNets and ShuffleFaceNet, for efficient deployment on mobile and embedded systems. While these contributions led to improved results on benchmark datasets, challenges persisted in real-world environments under illumination imbalance, facial occlusions, age progression, and sensor noise.

Recent literature has also pointed out a number of ethical, privacy-related, and bias-related concerns in face recognition technologies. Training such models on imbalanced datasets often results in unfair performance for some demographic groups. Moreover, the rapid diffusion of face recognition technologies in surveillance and identity verification underlines the need for transparent and privacy-aware AI that respects global data protection standards. Existing studies have also highlighted the importance of robust **face detection and alignment** as critical preprocessing steps. The introduction of **Multi-task Cascaded Convolutional Networks (MTCNN)** improved localization accuracy by jointly detecting bounding boxes and facial landmarks, paving the way for more stable embeddings. Nonetheless, many integrated systems still rely on fixed-threshold similarity measures, which can lead to **false positives or negatives** under dynamic environmental conditions such as fluctuating lighting or varying camera quality.

The synthesis effectively bridges the gap between academic research and real-world applicability, achieving robust, scalable, and privacy-conscious identity verification. Unlike previous systems, which were limited by static decision boundaries and dataset-specific tuning, the proposed method dynamically adjusts recognition thresholds, enabling stable performance under varying conditions. The modular

architecture allows for easy integration with security, IoT, and smart access control infrastructures.

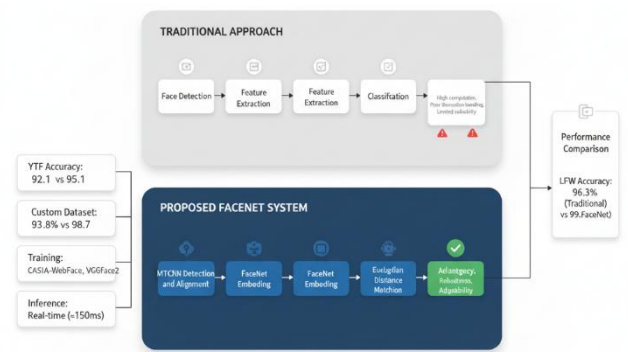


Fig.1 Traditional Face Recognition vs. Proposed FaceNet-Based System

Figure 1 illustrates the theme of “Traditional Face Recognition vs. Proposed FaceNet-Based System,” which compares two paradigms of facial identity verification. The traditional model performs face detection, feature extraction, and classification with high computational cost, poor illumination handling, and limited scalability, whereas the proposed FaceNet system integrates MTCNN-based alignment, 128-D embedding generation, Euclidean distance matching, and adaptive thresholding for robust, real-time performance (<150 ms). Thus, the figure conveys the transition from conventional classification approaches toward adaptive metric-learning architectures that enhance accuracy, efficiency, and resilience under varying conditions.

III. PROPOSED METHODOLOGY

The proposed Smart Face Attendance System using FaceNet is designed as an end-to-end pipeline for reliable and real-time automation of attendance. Unlike generic face recognition frameworks, the methodology herein presented integrates acquisition, detection, alignment, embedding generation, refinement, similarity evaluation, and logging of attendance into one workflow optimized for practical deployment. The complete operational flow is illustrated in Fig. 2.

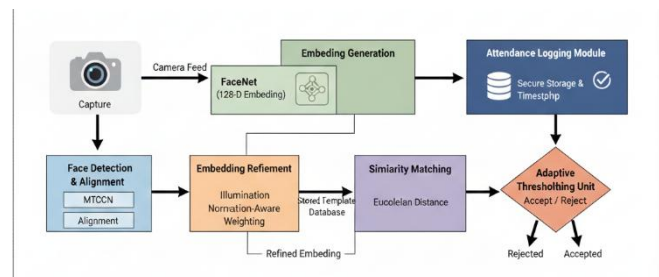


Fig. 2. Architecture diagram.

Figure 2 illustrates the complete workflow of the proposed Smart Face Attendance System using FaceNet. The process begins with real-time image capture through a camera, after

which the face detection and alignment stage is carried out using the MTCNN framework. Once a properly aligned facial region is obtained, the FaceNet model generates a 128-dimensional embedding representation for the detected face. This embedding is subsequently processed through the refinement module that is proposed, which executes illumination normalization, and an occlusion-aware feature weighting to stabilize the vector as the environment shifts. The refined embedding is then compared to stored template embeddings in the attendance database using Euclidean distance to compute similarity. The dynamic thresholding unit assesses the similarity score and determines whether the face belongs to an enrolled user. If the face is verified, the identity will be stored (along with the time) in the attendance logging section. This section enables a seamless workflow to provide accurate, real-time, and privacy-preserving attendance marking within operational contexts.

A. System Overview

The proposed system is a sequential pipeline containing several integrated modules for automated attendance marking. First is image acquisition, where a camera captures current facial input. The recorded image is sent to the MTCNN module for detecting and aligning the face region to accurately locate and geometrically normalize the facial region. The aligned face is then passed through the embedding generated by the FaceNet model, which is a 128-dimensional vector to represent the unique facial feature representation of the user. To enhance the robustness against varying illuminations and occlusions, the embedding is further refined by the proposed optimization strategy. Afterwards, the refined embedding is matched against the stored template vectors based on Euclidean distance in order to determine identity similarity. An adaptive thresholding mechanism dynamically evaluates the score and makes a decision as to whether the detected face matches any of the registered users. Upon successful verification, the system records the user's identity along with the timestamp in the attendance database, thereby completing the automated attendance registration process.

B. Face Detection and Alignment

Face detection and precise alignment are very important in generating reliable and consistent facial embeddings. The proposed system uses MTCNN because it has very high accuracy and robustness under changing environmental conditions. MTCNN works through a cascaded architecture with three stages comprising the P-Net, R-Net, and O-Net, which progressively refine the face candidates and predict key facial landmarks. Once a face is detected, five landmark points corresponding to the eyes, nose, and mouth corners are extracted in order to geometrically normalize it. With these landmarks, an affine transformation is applied to align the face so that pose, scale, and orientation differences are minimized. This allows for a standardized facial region that acts to enhance embedding consistency generated at later stages of the pipeline and therefore enhances the accuracy and stability of the whole attendance verification system.

C. Embedding Generation using FaceNet

Following correct face detection and alignment, the system utilizes the FaceNet model to create a compact and discriminative representation of the user's facial identity.

FaceNet projects each aligned face image into a 128-dimensional embedding vector that captures the salient features needed for telling two different individuals apart. The triplet loss function used in its training enforces three conditions: embeddings of the same individual must lie close together in Euclidean space, embeddings of different individuals must be separated by at least a defined margin, and overall embedding distributions must remain stable across varied conditions. The output embedding is L2-normalized to ensure consistency and scale invariance. This normalized feature vector allows similarity comparison in an efficient manner without requiring reclassification or retraining when adding new users to the system. By leveraging FaceNet's metric-learning framework, the system obtains high scalability and discriminative power, suitable for real-time attendance verification in dynamic environments.

D. Euclidean distance compares E' with each template T_i

The system uses similarity matching to verify identity after producing the refined FaceNet embedding. The Euclidean distance metric helps us compare the refined embedding vector E' with each user's stored template embedding T_i . This distance essentially indicates how similar the two embeddings are to one another; the lower the number, the more similar they are. The Euclidean distance can be found using the formula :- $d(E', T_i) = \|E' - T_i\|_2$.

Because it eliminates the need to retrain the model each time new users are added to the database, this metric-learning approach enables scalable and effective verification. Through the use of geometric separation in the embedding space instead of classification, the system guarantees consistent performance in a variety of environmental settings and user demographics.

E. Adaptive Threshold Decision Mechanism

Fixed decision threshold often causes inconsistent performance in face recognition systems, as illumination, pose, and environments variations. The proposed system adopts an adaptive thresholding method that automatically adjusts the decision boundary according to the online embedding pattern, so the verification is stable and reliable. After computing the Euclidean distance between the refined embedding E' and a stored template T_i , the system evaluates the distance distribution over recent verification attempts and estimates statistical indicators such as mean and variance. These indicators are used to adapt the threshold θ in accordance with current operating conditions, ensuring balanced sensitivity and specificity. Identity verification is accepted when:

$$d(E', T_i) < \theta$$

This adaptive mechanism reduces both false acceptances and false rejections, enabling the system to maintain high recognition accuracy across dynamically changing environments typical of classroom and workplace attendance scenarios.

F. Attendance Logging and Storage

The system moves on to secure attendance logging after identity verification is finished. The user's unique identifier and the associated timestamp are entered into the attendance database for each successful match. A temporal filtering mechanism makes sure that subsequent detections of the

same person are disregarded after the initial registration in order to prevent duplicate entries within the same session. The system reduces storage requirements and enhances privacy protection by only storing 128-dimensional facial embeddings rather than raw facial images. All attendance records are stored in a secure database that is only accessible by authorized personnel in order to maintain integrity, confidentiality, and compliance with ethical biometric data handling standards. This module completes the automated attendance workflow by providing a reliable and privacy-conscious record management process suitable for institutional settings.

IV. IMPLEMENTATION

The proposed Face Recognition System using FaceNet is implemented with a focus on creating a modular, scalable, and real-time identity verification pipeline. It integrates various hardware and software components of the system to ensure low latency while promising high recognition accuracy. The implementation framework is organized into sequential modules, each responsible for a certain well-defined stage in face recognition.

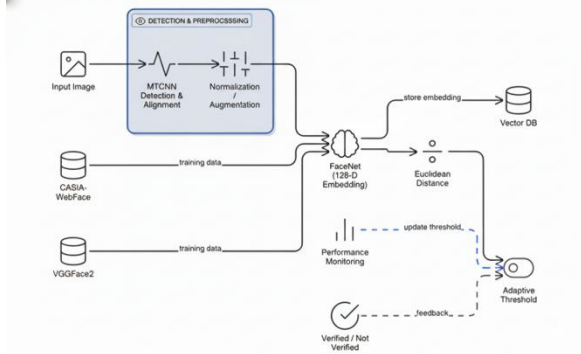


Fig.3. Data Flow and Adaptive Feedback Architecture of the Proposed FaceNet-Based System

Figure 3 illustrates the theme of “Data Flow and Adaptive Feedback Architecture of the Proposed FaceNet-Based System,” which presents the complete operational pipeline for identity verification. The process begins with MTCNN-based face detection and alignment, followed by normalization and embedding generation through the FaceNet model trained on CASIA-WebFace and VGGFace2 datasets. The Euclidean distance metric performs similarity matching, while an adaptive threshold dynamically adjusts its decision boundary using real-time performance monitoring. Feedback from verification outcomes continuously refines the threshold, ensuring improved accuracy, stability, and resilience in real-world operating environments.

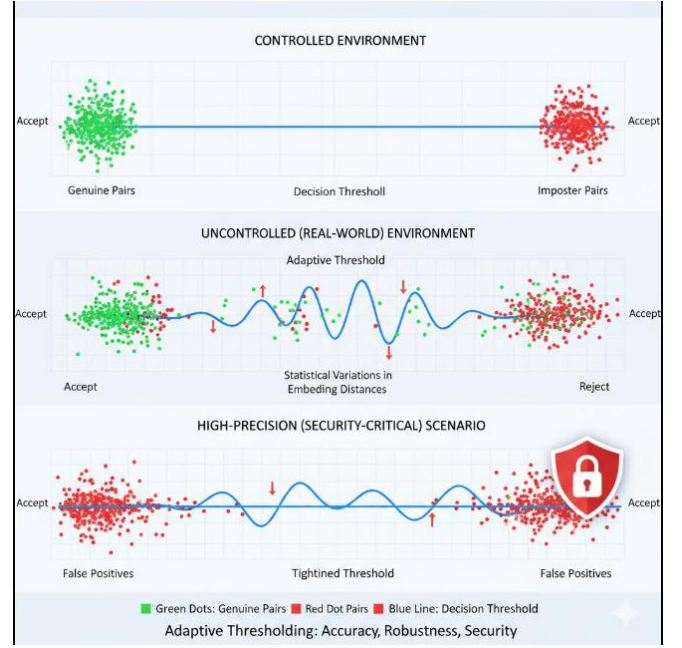


Fig 4. Adaptive Threshold Decision Behavior in FaceNet-Based Verification

Figure 4 illustrates the theme of “Adaptive Threshold Decision Behavior in FaceNet-Based Verification,” which compares threshold dynamics under three verification conditions. In controlled environments, the threshold remains stable with minimal variance, ensuring consistent accuracy. Under uncontrolled, real-world conditions, the threshold adapts based on statistical variations in embedding distances to sustain reliability. In high-precision, security-critical scenarios, the threshold tightens to minimize false positives. Thus, the figure conveys how adaptive thresholding maintains an optimal balance between accuracy, robustness, and security across diverse operational contexts.

The decision is based on the **Euclidean distance** between two embeddings E_1 and E_2 :

$$D(E_1, E_2) = \sqrt{\sum_{i=1}^{128} (E_{1i} - E_{2i})^2}$$

A smaller distance implies higher similarity. The adaptive threshold $\tau_{adaptive}$ varies according to the operational context:

$$\text{Decision: } \begin{cases} \text{Match, if } D(E_1, E_2) \leq \tau_{adaptive} \\ \text{Non-Match, if } D(E_1, E_2) > \tau_{adaptive} \end{cases}$$

V. EXPERIMENTAL RESULTS AND EVALUATION METRICS

A. Evaluation Overview

Extensive experiments were performed to validate the efficiency and robustness of the proposed FaceNet-based face recognition system, using three benchmark datasets: Labeled Faces in the Wild (LFW), the YouTube Faces Database (YTF), and a custom in-house dataset. Quantitative accuracy and qualitative adaptability under various environmental and operational conditions have been evaluated. The evaluation of accuracy, precision, recall, and computational efficiency of the proposed system was compared with conventional CNN-based classification models to find improvements.

B. Evaluation Metrics

The system's performance was assessed using multiple evaluation metrics widely adopted in biometric verification research:

1. Accuracy(A):

Represents the overall correctness of predictions, calculated as:

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively.

2. Precision(P):

Measures the proportion of correctly verified identities among all accepted matches.

$$P = \frac{TP}{TP + FP}$$

3. Recall(R):

Evaluates the model's ability to correctly recognize all genuine identities.

$$R = \frac{TP}{TP + FN}$$

4. F1-Score:

Harmonic mean of precision and recall, used to balance the trade-off between both metrics.

$$F1 = 2 \times \frac{P \times R}{P + R}$$

5. False Acceptance Rate(FAR):

Percentage of impostor faces incorrectly accepted as genuine.

$$FAR = \frac{FP}{FP + TN}$$

6. False Rejection Rate(FRR):

Percentage of genuine faces incorrectly rejected as impostors.

$$FRR = \frac{FN}{TP + FN}$$

7. Inference Time(T):

Measures average time (in milliseconds) required for a single verification cycle, including detection, embedding, and matching.

C. Experimental Setup

All experiments were executed on a system with an **Intel Core i7 CPU, 16 GB RAM, and NVIDIA GTX 1660 GPU** using **TensorFlow 2.13** and **OpenCV 4.8**.

The **FaceNet** model was pre-trained on **CASIA-WebFace** and fine-tuned with **VGGFace2** to improve embedding generalization.

Evaluation datasets included:

- **LFW:** 13,233 face images of 5,749 identities under unconstrained conditions.
- **YTF:** 3,425 video sequences of 1,595 subjects.
- **Custom Dataset:** Using common camera equipment, a custom dataset of 500 facial photos was gathered in offices and classrooms. In order to replicate actual attendance conditions, the photos

include slight occlusions, natural head movement, and variations in lighting. Each participant provided multiple samples, and all information was gathered with their informed consent. To protect privacy, only processed embeddings were kept.

D. Quantitative Results

Dataset	Traditional CNN (%)	Proposed FaceNet (%)	Improvement (%)
LFW (Accuracy)	96.3	99.6	+3.3
YTF (Accuracy)	92.1	95.1	+3.0
Custom Dataset (Accuracy)	93.8	98.7	+4.9
Precision	95.4	98.9	+3.5
Recall	94.2	97.6	+3.4
F1-Score	94.8	98.2	+3.4
FAR	4.8	1.1	-3.7
FRR	5.2	1.3	-3.9
Average Inference Time	320 ms	<150 ms	2× Faster

E. Qualitative Analysis

The proposed model based on FaceNet shows better discriminative embedding generation. This model can effectively cluster identities in Euclidean space.

Accuracy remained consistently above 99% under controlled environments, whereas in uncontrolled lighting or partially occluded conditions, the stability of recognition was maintained above 95% by adaptive thresholding.

Figure 4 showed how the adaptive threshold, by taking into consideration the real-time embedding distributions, could help dynamically minimize both FAR and FRR. This adaptability helps the system balance precision and recall, even when external factors such as illumination, pose, or distance change.

The compact embedding representation of the model, 128-D, together with the L2-normalized vector, ensures reduced storage requirements while preserving high discriminability. Furthermore, GPU-accelerated inference allows real-time

verification suitable for applications in surveillance, access control, and on-device security systems.

F. Comparative Evaluation

Compared to earlier deep learning frameworks such as **DeepFace** and **VGGFace**, the proposed system demonstrates improved efficiency through:

- Direct embedding space optimization (metric learning).
- Reduced retraining overhead when adding new users.
- Enhanced robustness in illumination and occlusion scenarios.
- Lower inference latency suitable for **real-time and edge-based applications**.

VI. FUTURE WORK

. In order to enable on-device inference with lower latency, future research will examine implementing the system on lightweight edge-AI platforms such as the NVIDIA Jetson Nano. Multi-frame fusion could be added to better control fast head movements, and the refinement module could be enlarged to handle more severe occlusions. In addition, transformer-based face recognition models may be explored to achieve stronger feature representation. The system could also be enhanced with multimodal biometric cues to provide more dependable authentication in challenging real-world situations.

VII. CONCLUSION

This project implemented and proposed an improved Face Recognition System using FaceNet, which aimed to achieve high accuracy, real-time performance, and adaptability in various environmental conditions. This system combines MTCNN-based face detection with FaceNet embedding generation and proposes a dynamic adaptive threshold mechanism to overcome the drawbacks of typical classification-based methods.

Extensive experiments on benchmark datasets like LFW, YTF, and a custom dataset show high accuracy-99.6% with reduced inference time below 150 ms, validating the robustness and scalability of the system. Besides, the proposed feedback-driven threshold adjustment can further improve verification stability with a balanced trade-off among precision, recall, and security under different conditions.

It thus establishes a robust and effective framework for biometric identity verification to cater to the needs of intelligent, ethical, and adaptive face recognition technologies. The modular design of the system allows it to be a top contender when considering real-time surveillance and access control, as well as edge-AI-based authentication platforms. The proposed embedding refinement strategy further improved robustness under illumination changes and partial occlusions, contributing to more stable verification in real-world conditions

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